

Unity GIS Tech 2018-2022



Terrain Streaming

Powerful Terrain Streaming Tools for Unity Engine

Documentation v2.0

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➤ Overview

Description

Terrain Streaming is a Unity plugin designed to generate a huge geo-referenced real world close to the reality environments.

Thanks to multiple synchronized functions and sectors intersection, the plugin is able to load the real world data (DEM-Raster-Vector) in a smart behavior to dynamically load and unload terrains around the player.

Using the plugin gives us the ability to generate high terrains resolution and easily manage arbitrary amounts of terrain in real-world terrain simulation and open-world video games all in offline mode.

This Asset include two parts, first is GIS Data Downloader which is the base part to download and generate GIS tiles, the second is the runtime terrain streaming which is responsible to load-unload data dynamically around the player.

Once the data is downloaded via the GIS data downloader, Terrain Streaming can easily load them from StreamingAsset folder or from any location in the hard disk.

All Generated Terrains are geo-referenced automatically to WGS84, that give us the ability to Get/Set any Vector3, Transform or GameObject position from and to Unity World Space <----> Real World Coordinates.

*** FEATURES ***

GIS Data Downloader

- Accurate area Bounds selection by using custom interactive real world map;
- Download easily almost of Geographic Information System (GIS) data (DEM-Raster-Vector) ;
- Download high-resolution satellite imagery and maps;
- Generate Heightmaps and Textures tiles for streaming purpose;
- Ability to generate a geo-referenced raster tiles by creating a Tab files to import them into any GIS software;
- Save/Load Area bounds to file for future using;
- Ability to set a custom tiles count;
- Unlimited number of generated terrains;
- Ability to avoid overwriting downloaded files to save downloading time;
- Auto calculation to Get the best tiles account to optimize streaming;
- Customized GUI interface.
- Elevation providers:
 - SRTM v4.1 resolution of 90 meters per pixel;
 - SRTM30 resolution of 30 meters per pixel.
- Raster providers:
 - ArcGIS Maps;
 - Mapbox Maps;
 - Virtual Earth (Bing Maps).

- Vector providers:
 - OpenStreetMap.

Runtime Terrain Streaming

- Generate terrains at runtime terrains;
- Save memory and FPS of your game by loading only a small part of your area;
- Customized GUI for Terrain System preferences;
- Ability to scale up/down generated world;
- Ability to modify terrains preferences at runtime;
- Load and unload dynamically terrains around the player;
- Customize the size of the loaded zone area;
- Customize the player intersection mode (FOV - Rectangle – Circle) ;
- Generate Random Trees and Grass Randomly or from Vector File Data;
- Generate Roads Vector File Data;
- Customize terrains material for HD pipeline purpose;
- Get-Set Coordinates in multiple projections;
- Full source code included.

***** SUPPORT *****

- API Compatibility Level .Net 4.x
- Compatible with Unity's terrain system;
- Easy to use with video & documentation;
- Tested with Unity 2019,2020, 2021;
- Active support through emails, forum posts and discord;
- The package contains the complete source code without dependencies.

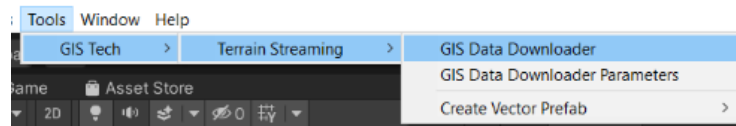
➤ GIS Data Downloader

Quick start

From menu click on «Tools /GIS Tech/ Terrain Streaming/ GIS Data Downloader », to open GIS Data Downloader window.

Enter the coordinates of the wanted area in decimal, configure the fields and click «Download Data».

GIS Data Downloader will automatically download all necessary GIS data to generate terrain tiles for the streaming system.

A screenshot of the 'GIS Data Downloader' window in Unity. The window has a title bar with 'GIS Data Downloader' and an 'Inspector' tab. It is divided into two main sections: 'Coordinates' and 'Download Options'.

The 'Coordinates' section includes:
- A 'Zone Name' field with the value 'Mojave_Desert'.
- 'Upper-Left' coordinates: Latitude '35.318788' and Longitude '-116.128682'.
- 'Down-Right' coordinates: Latitude '34.9744746' and Longitude '-115.5969071964'.
- Action buttons: 'Open Map Helper', 'Paste Coordinates', 'Zone Info + Optimized Input', 'Save Zone', and 'Load Zone'.

The 'Download Options' section includes:
- A 'Select Location' button.
- A 'Terrain Path' field with the value 'E:/Terrain Streaming Examples/Mojave_Desert'.
- Four dropdown menus for data types, all set to 'Disable': 'Download DEM Data', 'Download Raster Data', 'Download Vector Data', and 'Download Terrain Horizon'.
- A 'Grid Mode' dropdown menu set to 'Auto'.
- Two dropdown menus for preview generation: 'Generate Preview Raster' (set to 'Enable') and 'Generate World Preview' (set to 'Disable').
- A large 'Download Data' button at the bottom.

Description

Fields Description

We have added a tooltip to each field in GIS Data Downloader Windows (Editor-Runtime) to help users to understand how each element works.

1- Zone Name

- "Zone Name": Description of the current Area.

2- Coordinates Bounds

- The coordinates of the top-left and bottom-right corners of the area in decimal Latitude-Longitude.

3- Buttons

- Open Map Helper : Open an online maps to easily select the wanted zone bounds.
- Paste Coordinates: Inserts the coordinates copied in clipboard from the map helper.
- Zone Info + Optimized Input: Calculate and Debug the optimized inputs + Information about that area (Length + width) in [Km].
- Save Zone: Save Area Bounds to file.
- Load Zone: Load Area Bounds.

4- Download Path

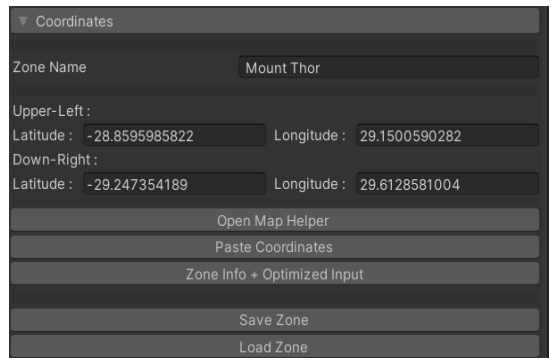
- Terrain Path: Select a directory to store downloaded data, by clicking on 'Select Location' button.

5- DEM Downloader

- Download DEM Data: Enable/Disable DEM Downloading.
- DEM Source: Select the elevation provider source (SRMT90/SRTM30), SRTM30 is recommended.
- Heightmap Resolution: The pixel resolution of the generated Terrain's Heightmap.
- Username + Password : Identification for Earthdata to get access to SRTM30 elevation.

6- Raster Downloader

- Download Raster Data: Enable/Disable Raster Downloading.
- DEM Source: Select the textures provider source, ArcGIS is recommended.
- Map Type : Aerial/Street/Terrain ... etc.
- Key : Provider API Key (for mapbox and bings).
- Zoom Level: Maps Zoom level.



Coordinates

Zone Name: Mount Thor

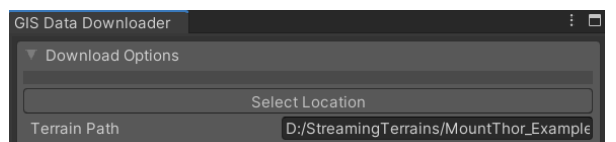
Upper-Left:
Latitude: -28.8595985822 Longitude: 29.1500590282

Down-Right:
Latitude: -29.247354189 Longitude: 29.6128581004

Buttons: Open Map Helper, Paste Coordinates, Zone Info + Optimized Input, Save Zone, Load Zone



Interactive Map Helper



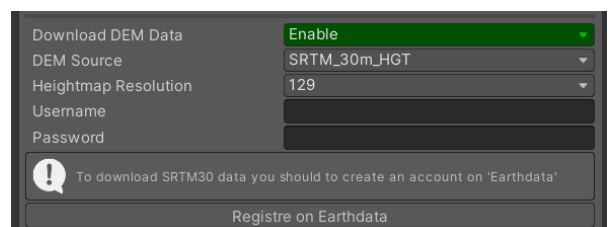
GIS Data Downloader

Download Options

Select Location

Terrain Path: D:/StreamingTerrains/MountThor_Example

GIS Data Directory



Download DEM Data: Enable

DEM Source: SRTM_30m_HGT

Heightmap Resolution: 129

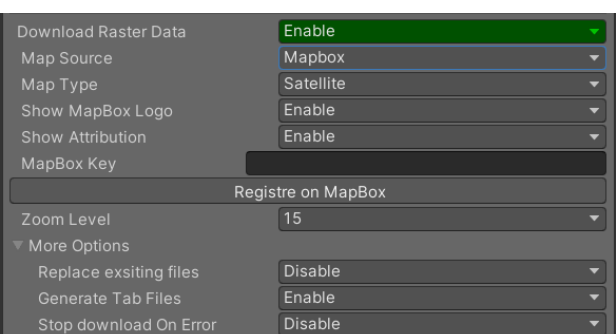
Username: [empty]

Password: [empty]

To download SRTM30 data you should to create an account on 'Earthdata'

Buttons: Register on Earthdata

DEM Downloader



Download Raster Data: Enable

Map Source: Mapbox

Map Type: Satellite

Show MapBox Logo: Enable

Show Attribution: Enable

MapBox Key: [empty]

Buttons: Register on MapBox

Zoom Level: 15

More Options:

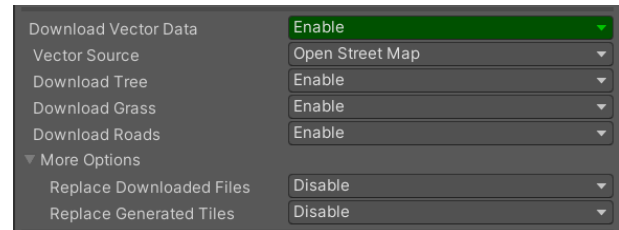
- Replace exsiting files: Disable
- Generate Tab Files: Enable
- Stop download On Error: Disable

- More Options :
Replace Existing Files: Disable this option to avoid overwriting and replacing downloaded files.
Generate Tab File : Enable it to generate tab which helps to import Raster tiles to any GIS Software for future using.
Stop download On Error : abort downloading when a server not responds.

7- Vector Downloader

- Download Vector Data: Enable/Disable Vector Downloading.
- Download Tree: Enable it to download Trees vector data.
- Download Grass: Enable it to download Grass vector data.
- Download Roads : Enable it to download Roads vector data.
- Download Building : //in Future Version

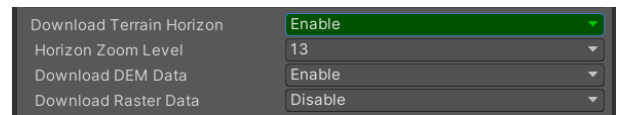
7



8- Terrain Horizon

- Download Terrain Horizon: Use this option to enable downloading DEM + Raster data to get a large terrain vision and a great background for your scene.

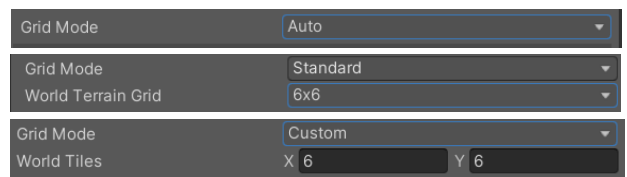
8



9- Grid Mode

- Grid Mode: Use this option to define the number of terrain tiles, in Auto GDD will calculate the best and optimized number of tiles for your world. Standard and custom used to set manually the number of tiles.

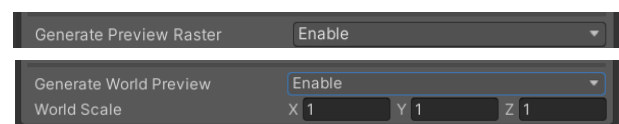
9



10- Generate Preview Zone

- Enable this option to generate a geo-referenced image, which can be imported into any GIS software.

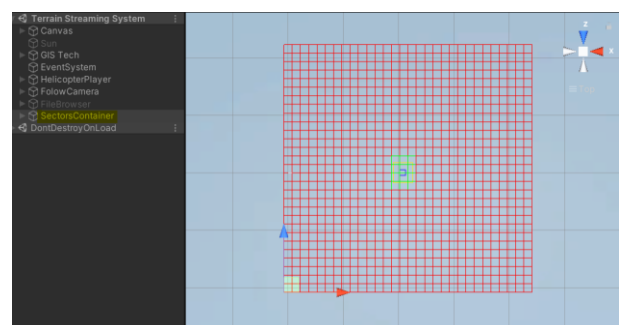
10



11- Preview the world

- Generate World Preview: Enable this option generate preview the your world before using runtime Streaming terrain, it's based on small sectors shown in the editor mode, Once you hit generate a new gameobject add to your scene called "SectorsContainer" it's the main sector container for your world, click on it and check your world metadata.(Check pictures in the next page)
- World Scale: Specifies the terrain scale factor in three dimensions (if world is large with 1 value you can set small float value like 0.5f - 0.1f - 0.01f".(Check pictures in the next page)

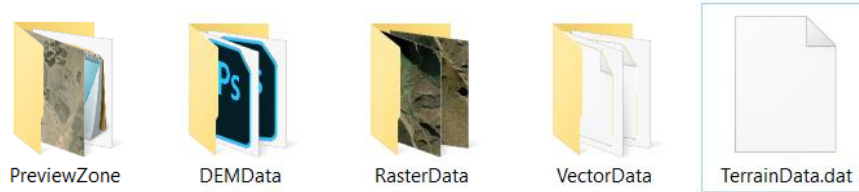
11



World preview Sectors

Downloaded Data structure

Once GIS data downloader is finished from downloading, check your folder and you will get the data ordered to many folders (DEM, Raster and Vector) + TerrainData.dat which will be called by terrain streaming system.



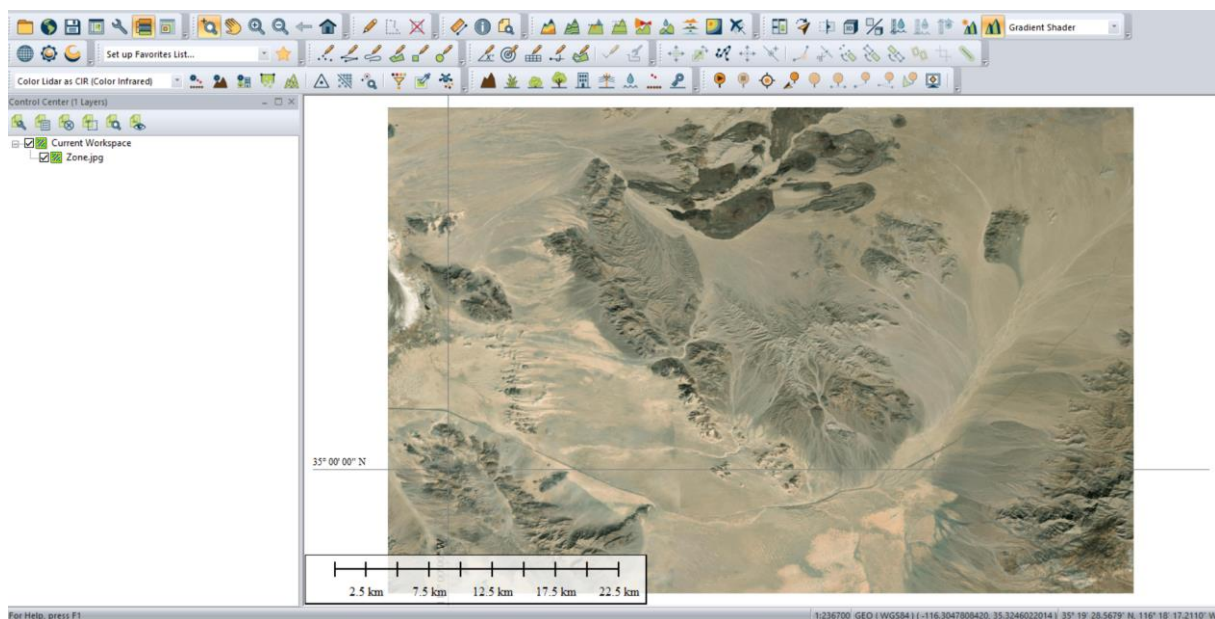
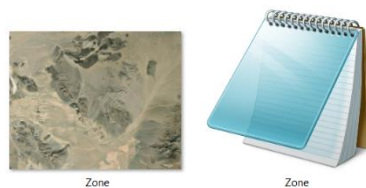
Link for Some Data Downloaded via GIS Data Downloader for streaming purpose:

Example 01: Texas Desert File Size 1.8GB [Download](#)

Example 02: Mojave Desert File Size 95 MB [Download](#)

Preview Zone:

Contains Geo-referenced raster of the whole zone, it is easy to import it directly to any GIS application (For analyzing purposes).



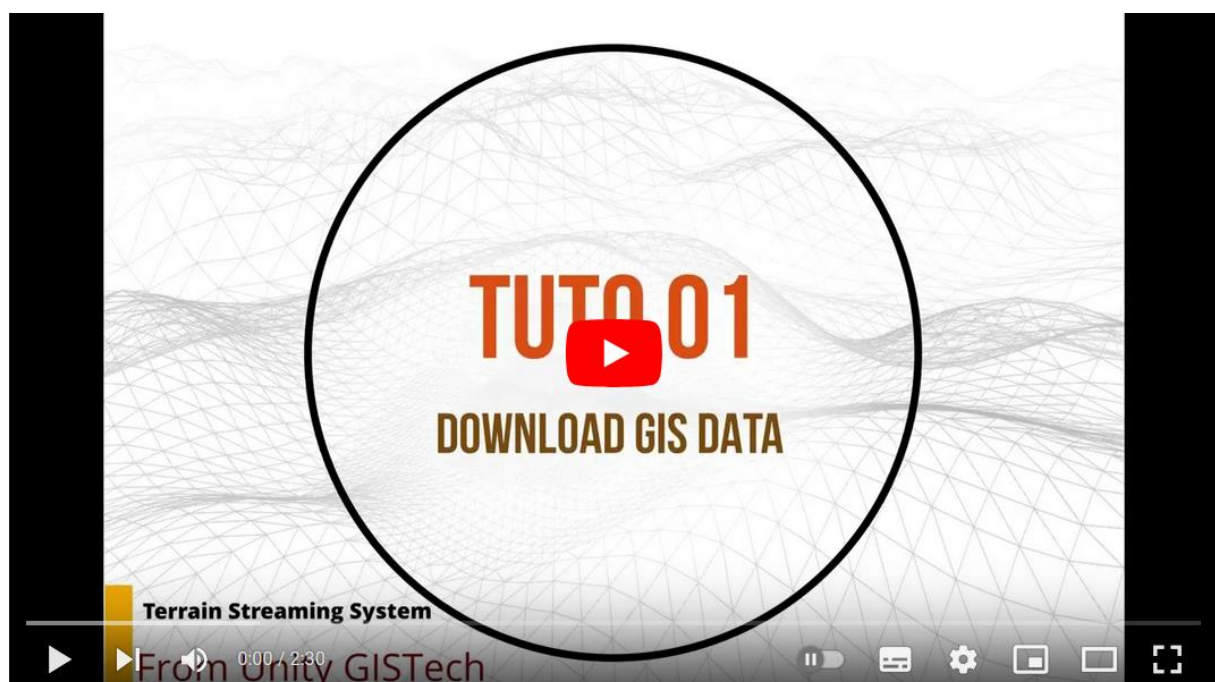
Zone Loaded in GIS software (Globalmapper)

Providers

GIS Data downloader use:

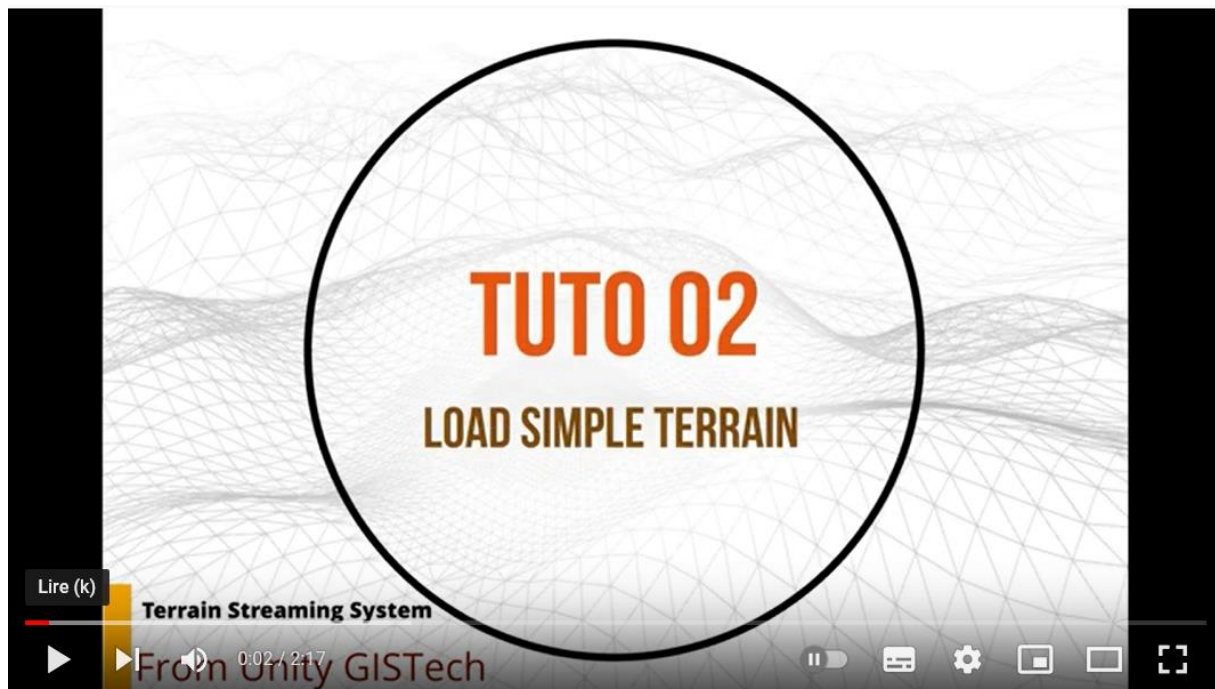
- For DEM Data:
 - SRTM 90m v4.1. Accuracy of 3 arcseconds (90 meters) (Without any key/token needed).
 - SRTM 30 m in HGT format from “Virtual Earth” , needed an identification on the site.
- For Raster Data:
 - ArcGIS Maps (No token needed).
 - MapBox (with API token).
 - Bingmaps (with API token).
- For Vector Data:
 - From Streetmap osm vector data.

For More watch that video tutorial: [Video Link](#)

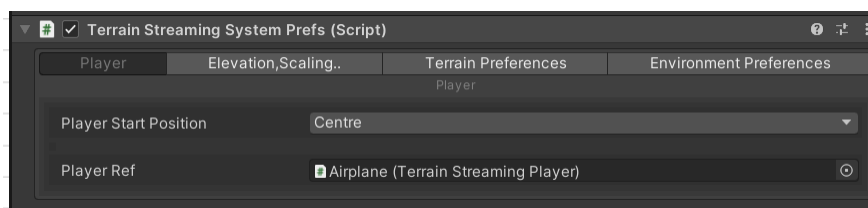


➤ Runtime Terrain Streaming

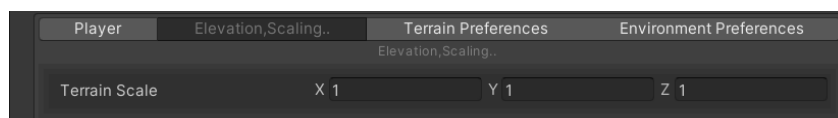
This Video tutorial explain how to use Runtime Terrain Streaming to Generate and Stream download data : [Video Link](#)



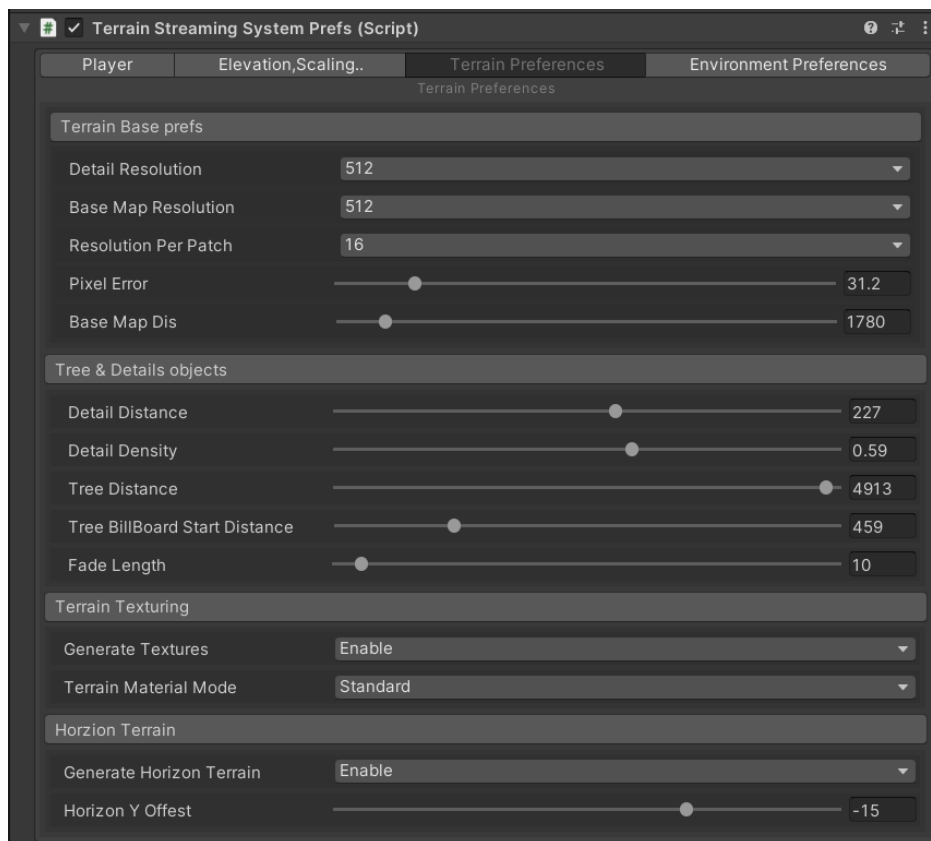
- 1- **TerrainStramingSystemPrefs.cs** : Contains all Terrain Streaming parameters and terrain preferences.



- Player Start Position : Set the Start Player Position (the centre of the world / Custom by setting Latitude and longitude position).
- Player Ref : Reference to the player script .



- Terrain Scale: Scale up or down you world (this is float value se we can set any values under 1f).



Terrain Base Preferences:

- Detail Resolution: The number of cells available for placing details onto the Terrain tile used to controls grass and detail meshes. Lower you set this number performance will be better.
- Base Map Resolution: Resolution of the composite texture used on the terrain when viewed from a distance greater than the Basemap Distance;
- Resolution Per Patch : The number of cells in a single patch (mesh), recommended value is 16 for very large detail object distance;
- Pixel Error: The accuracy of the mapping between Terrain maps and generated Terrain. Higher values indicate lower accuracy, but with lower rendering overhead
- Base map Distance: The maximum distance at which Unity displays Terrain Textures at full resolution, this improves performance for far away patches.

Tree & Details Objects:

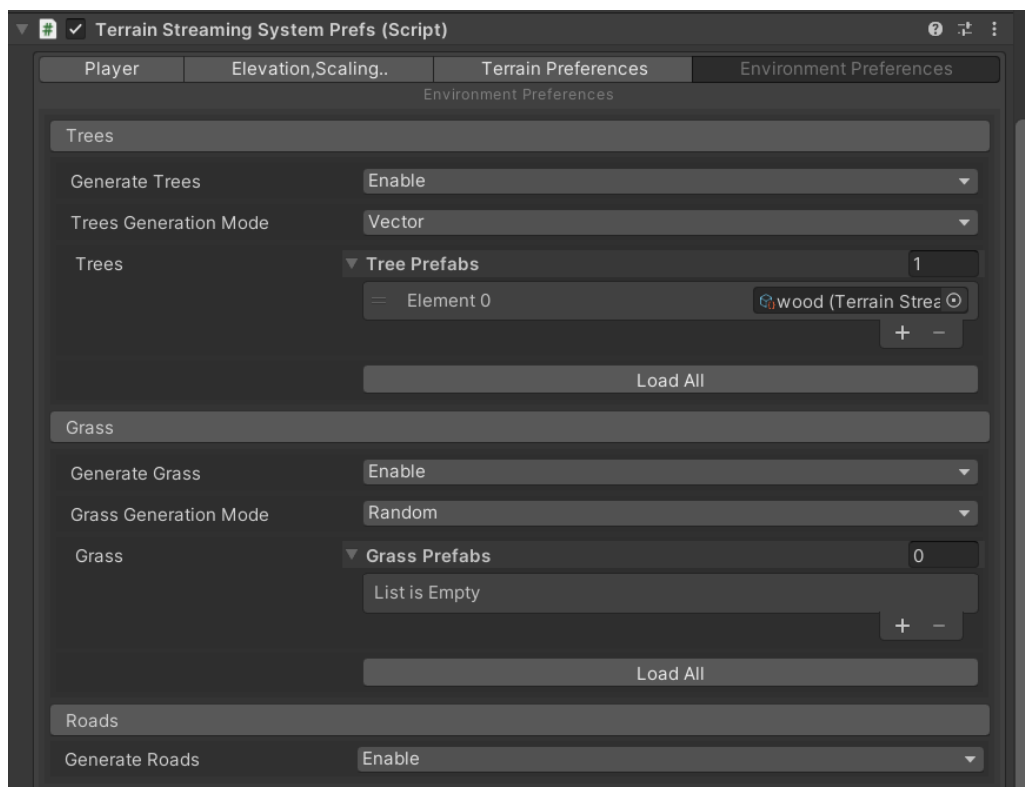
- Detail Distance: The distance from the camera beyond which details are culled.
- Detail Density: The number of detail/grass objects in a given unit of area. Set this value lower to reduce rendering overhead.
- Tree Distance: The distance from the camera beyond which trees are culled.
- Billboard Start: The distance from the camera at which Billboard images replace 3D Tree objects.
- Fade Length: The distance over which Trees transition between 3D objects and Billboards.

Terrain Texturing:

- Generate Textures : Enable/Disable generating textures during streaming terrains.
- Terrain Material Mode: Custom used when we want to change to main terrain material, used in case of HDRP.

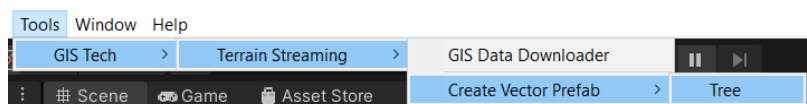
Horizon Terrain:

- Generate Horizon Terrain : Enable/Disable generating horizon terrain (Horizon data must be downloaded before enable this option) .
- Horizon Y Offset: Horizon Container Y offset, it must be negative value to make the Horizon Container under the Streamed terrain container.

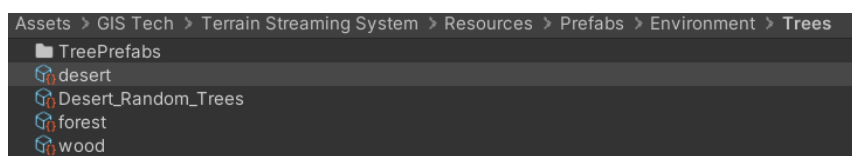


- Generate Trees : Enable/Disable trees generation.
- Generation Mode: Random Trees/ Vector: Generate trees randomly or from OSM vector data.

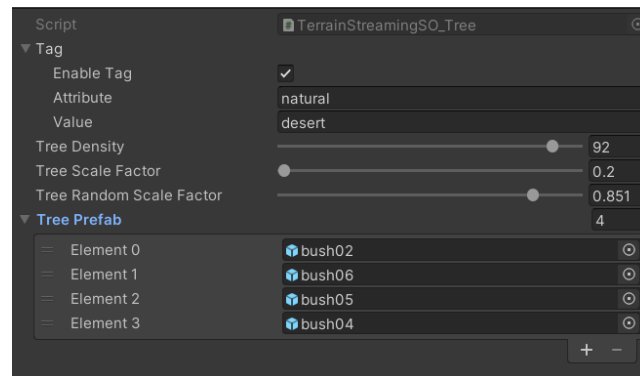
To Create Tree prefab:



Created Tree prefabs scriptableobject:



Tree ScriptableObject parameters :



Tree Tag: This parameter define that the current prefab will be loaded for any polygon has the same tags ex (Attribute : natural, Value : desert).

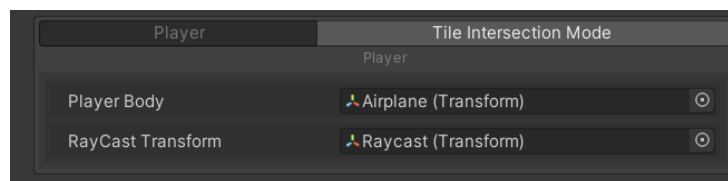
Tree Density: Tree Density controls the average number of Trees painted onto the loaded area defined by vector polygon size.

Tree Scale Factor: This will increase the size of all the trees set to the Tree Prefabs, only for that model that already defined by Tree Type.

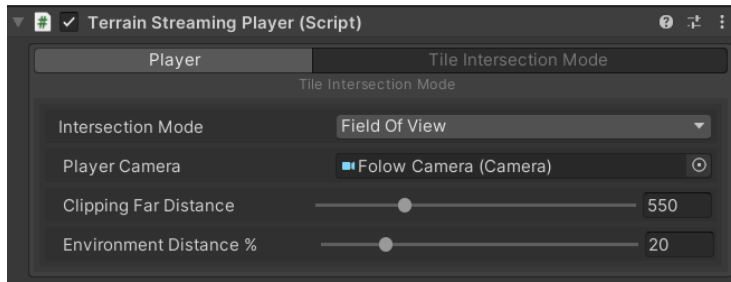
Tree Random Scale Factor: Give some realistic by changing randomly the size of trees in the same zone.

Tree Prefab: A list of tree prefabs that will generated in the defined zone.

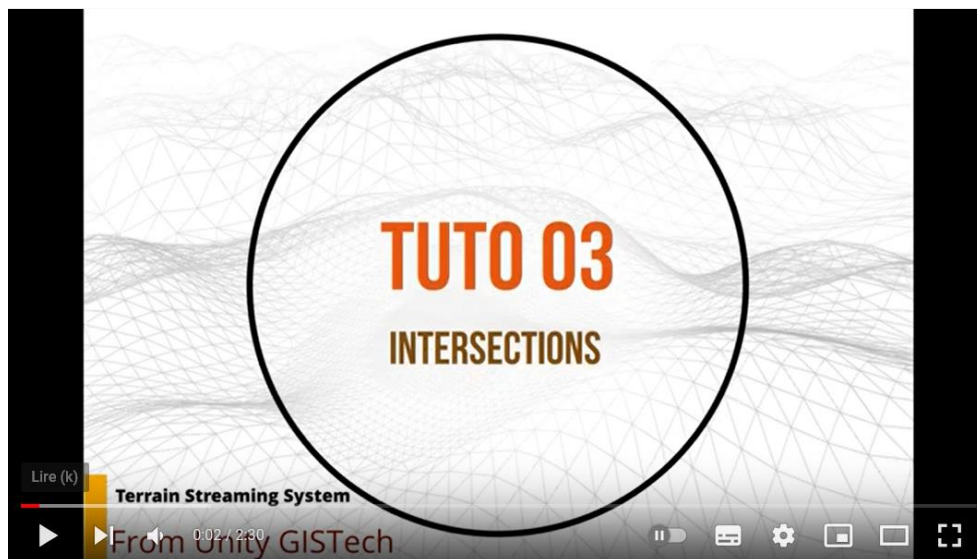
2- TerrainStramingSystemPlayer.cs : this script must be attached to the player.



- PlayerBody : Transform contains the Player Body.
- RayCast Transform : Used to raycast terrains (Raycast output).



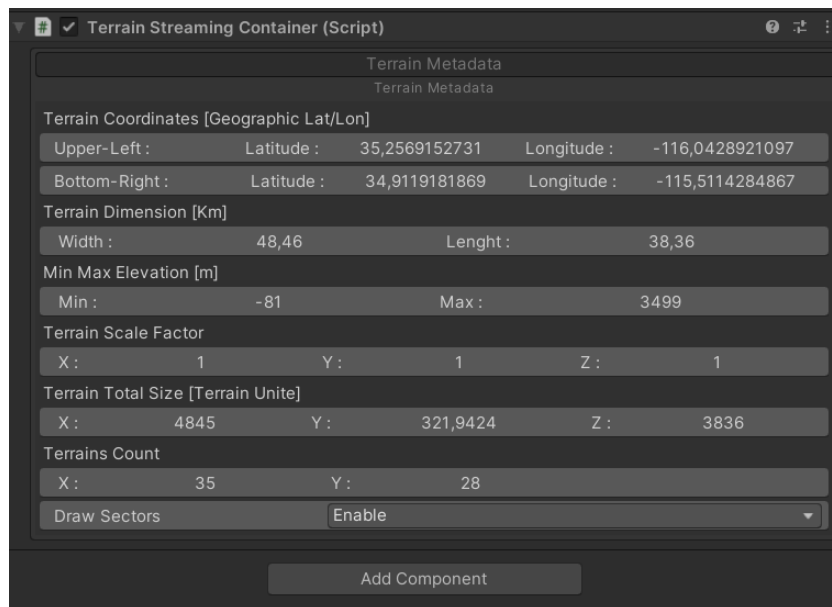
This Video tutorial explain TSS use tiles intersection to generate tiles around the player :
[Video Link](#)



Terrain Container Parameters

Terrain Metadata

Once the terrain is generated, we can get access to metadata panel from the inspector.



Terrain Metadata

Draw Sectors: used to enable/Disable sectors viewing in the editor mode.

Terrain Streaming Scripting

[SimpleTerrainStreaming.cs](#)



GIS Tech
UnityGISTech



➤ Supported projections

- Geographic Latitude Longitude (Decimal Degree, Deg Min Sec).
- UTM (Meter, Military Grid Reference).
- Lambert.

➤ Support

For any issues please contact us we will answer in 24 h maximum.

Email:

GISTech2008@Gmail.com

Forum Support page:

<https://forum.unity.com/threads/terrain-streaming-system.1133353/>

Discord Chanel

<https://discord.com/invite/3FhNP7YR54>

➤ Supported Platforms

The Current version support only Standalone for Mac, PC and Linux.

Other platforms will be supported in the future version.

➤ Links

YouTube page :

[GISTechTube](#)

Terrain Streaming Asset (To rate and review):

[Terrain Streaming System](#)